CO5_AT2 PSEUDOCODE WRITING TASK

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1. Design and implement a Backtracking algorithm to find the maximum independent set in a graph. The input is a graph represented using an adjacency matrix, and the output is the largest set of vertices such that no two selected vertices are directly connected by an edge. Use recursion and pruning to eliminate invalid vertex combinations efficiently.

CODE:

```
def maximum_independent_set(graph, n):
    def backtrack(vertex, current_set):

        # Pruning: remaining vertices cannot produce a larger set
        if len(current_set) + (n - vertex) <= len(max_set):
            return

        # Include the vertex if it is safe
        if is_safe(vertex, current_set, graph):
            current_set.append(vertex)
            backtrack(vertex + 1, current_set)
            current_set.pop()

        # Exclude the vertex
        backtrack(vertex + 1, current_set)

    backtrack(0, [])
    return max_set


# Adjacency matrix of the graph
graph = [
```

OUTPUT:

Output

```
Maximum Independent Set: [0, 3]
Number of vertices: 2

=== Code Execution Successful ===
```